Support Vector Machine (SVM) and Kernel Methods

Lecture 06

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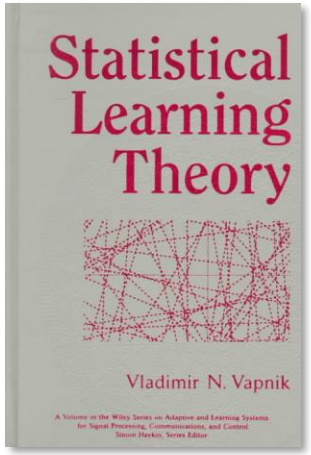
Outline

- **Support Vector Machine (SVM)**
  - history, main idea, mathematical derivation
- Soft-margin SVM
- Kernel methods

Support Vector Machine (SVM): history

- **理论基础（1960s）**
  - Vladimir Vapnik 和 Alexey Chervonenkis 提出了 VC维（Vapnik-Chervonenkis Dimension），为统计学习理论奠定基础，强调了模型复杂性与泛化能力的关系。
  - 这一理论后来成为SVM结构风险最小化(Structural Risk Minimization, SRM) 的核心依据。
- **最大间隔分类器的提出（1992年）**
  - 首次提出最大间隔超平面（Maximum-Margin Hyperplane） 的概念，解决线性可分问题。
  - 引入对偶优化问题，为后续核方法铺路。
- **现代SVM的正式成型（1995年）**
  - 核技巧（Kernel Trick）：通过非线性核函数（如高斯核、多项式核）将线性SVM推广到非线性分类。
  - 软间隔（Soft Margin）：允许部分样本违反间隔约束，提升对噪声数据的鲁棒性。



A training algorithm for optimal margin classifiers
BE Boser, IM Guyon, VN Vapnik - … of the fifth annual workshop on …, 1992 - dl.acm.org
… **ALGORITHM** In this Section, we highlight some important aspects of the **optimal margin training algorithm**… The description is split into a discussion of the qualities of the resulting **classifier**…
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Support-vector networks
C Cortes, V Vapnik - Machine learning, 1995 - Springer
… We also compare the performance of the **support-vector network** to various classical learning algorithms that all took part in a benchmark study of Optical Character Recognition. …
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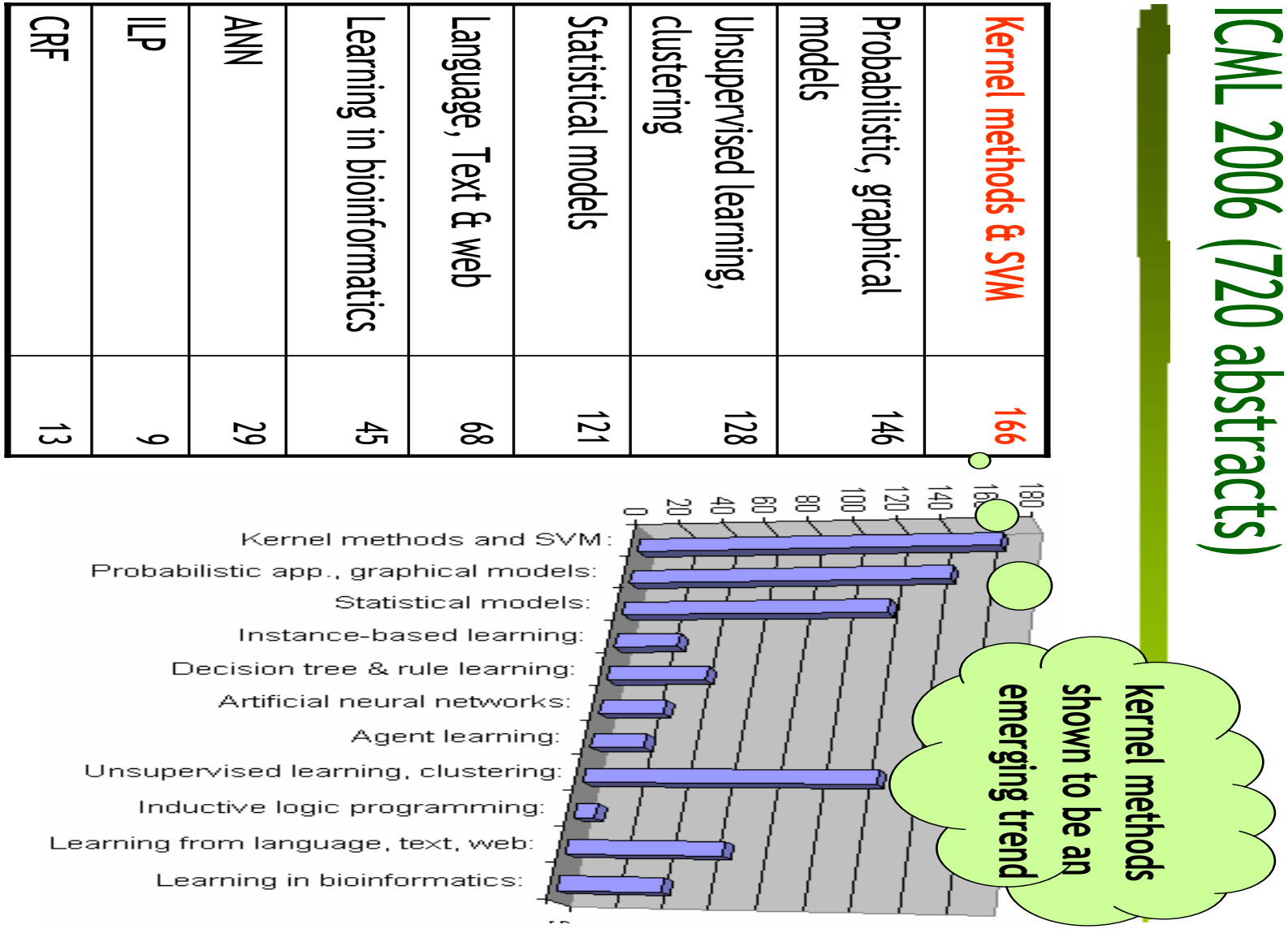
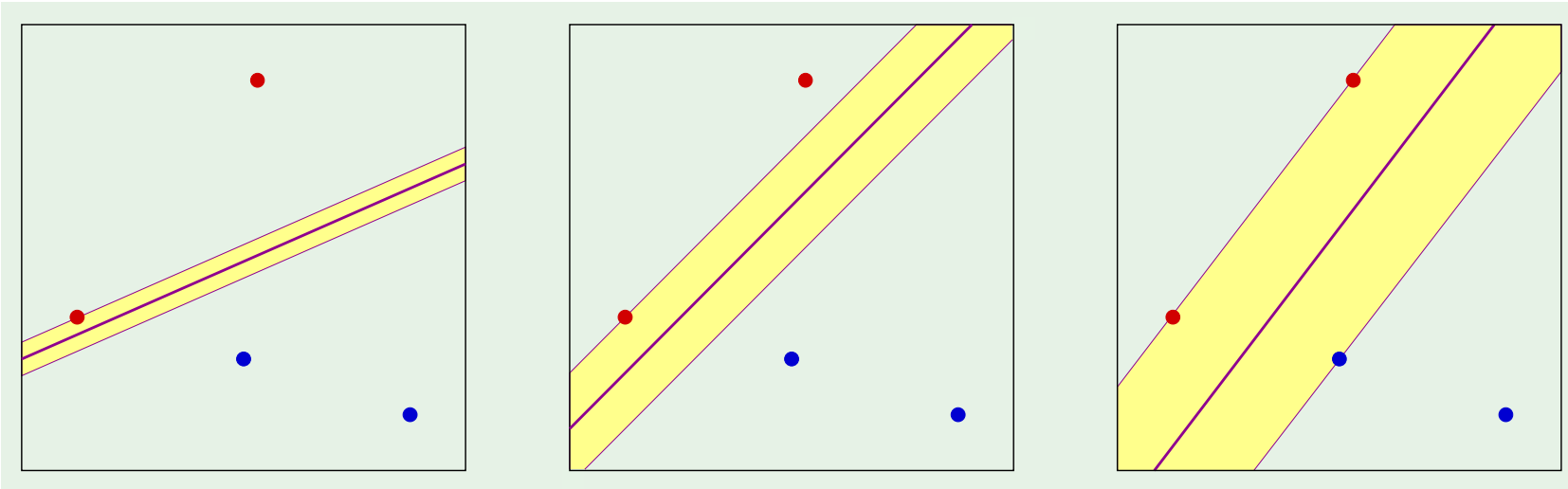
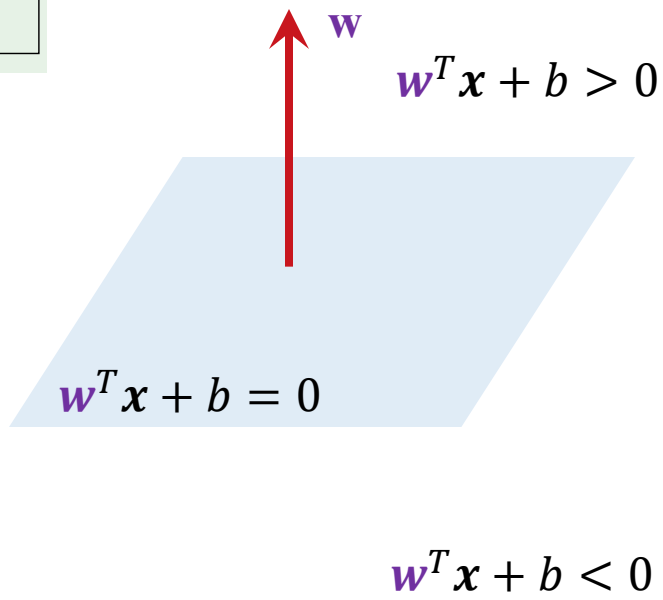
Very popular around 2006



Kernel methods & SVM	166
Probabilistic, graphical models	146
Unsupervised learning, clustering	128
Statistical models	121
Language, Text & web	68
Learning in bioinformatics	45
ANN	29
ILP	9
CRF	13

Which is a better linear separation





- Why is bigger margin better?
- Which hyperplane maximizes the margin?

Finding $\mathbf{w}$ with large margin

Let $\mathbf{x}_n$ be the nearest data point to the plane $\mathbf{w}^\intercal\mathbf{x} = 0$. How far is it?

2 preliminary technicalities:

1. **Normalize $\mathbf{w}$:**

   $$|\mathbf{w}^\intercal\mathbf{x}_n| = 1$$

2. **Pull out w_0:**

   $\mathbf{w} = (w_1, \cdots, w_d)$ apart from b

   The plane is now $\boxed{\mathbf{w}^\intercal\mathbf{x} + b = 0}$ (no x_0)

Computing the distance

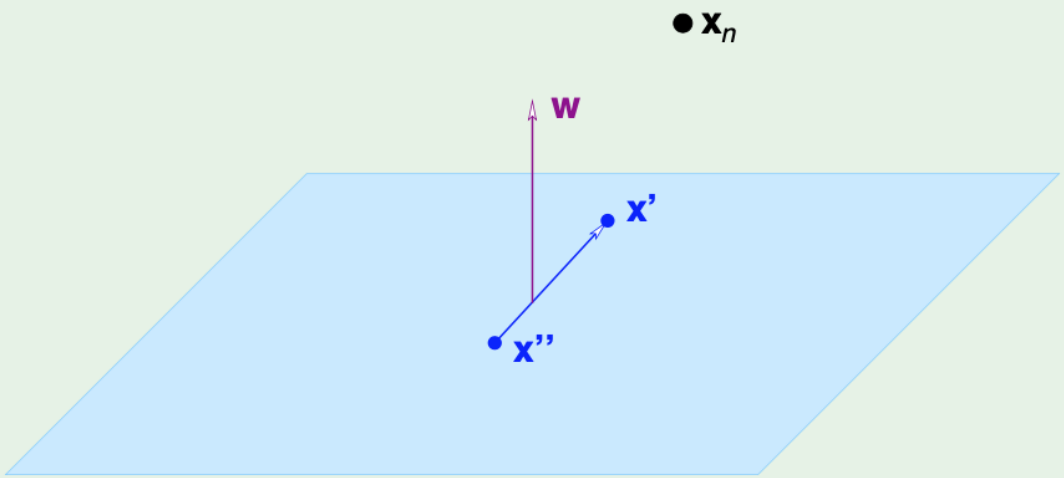
The distance between $\mathbf{x}_n$ and the plane $\mathbf{w}^\intercal\mathbf{x} + b = 0$ where $|\mathbf{w}^\intercal\mathbf{x}_n + b| = 1$

The vector $\mathbf{w}$ is $\perp$ to the plane in the $\mathcal{X}$ space:

Take $\mathbf{x}'$ and $\mathbf{x}''$ on the plane

$$\mathbf{w}^\intercal\mathbf{x}' + b = 0 \quad \text{and} \quad \mathbf{w}^\intercal\mathbf{x}'' + b = 0$$

$$\Longrightarrow \quad \mathbf{w}^\intercal(\mathbf{x}' - \mathbf{x}'') = 0$$

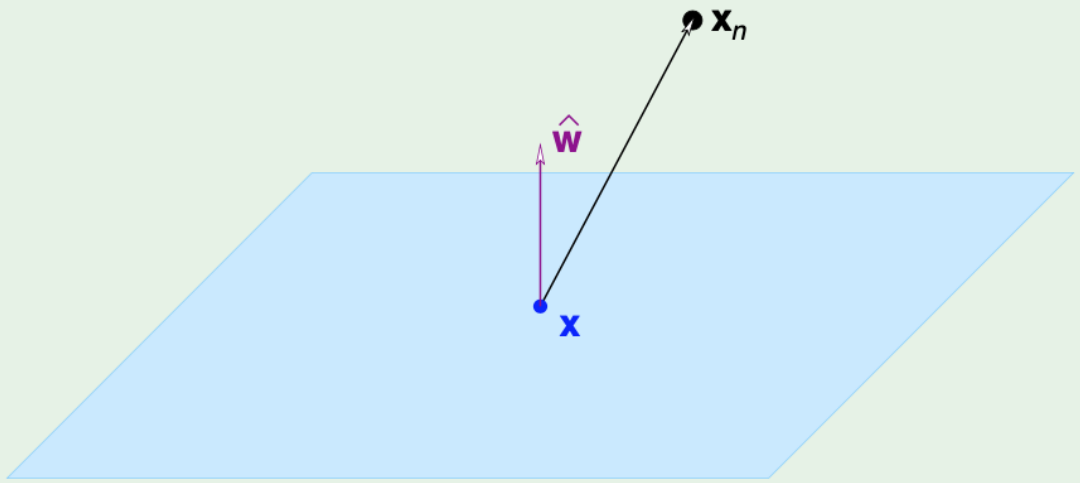


The distance from x_n to the plane

Distance between $\mathbf{x}_n$ and the plane:

Take any point $\mathbf{x}$ on the plane

Projection of $\mathbf{x}_n - \mathbf{x}$ on $\mathbf{w}$



$$\hat{\mathbf{w}} = \frac{\mathbf{w}}{\|\mathbf{w}\|} \implies \text{distance} = \left|\hat{\mathbf{w}}^\top(\mathbf{x}_n - \mathbf{x})\right|$$

$$\text{distance} = \frac{1}{\|\mathbf{w}\|}\left|\mathbf{w}^\top\mathbf{x}_n - \mathbf{w}^\top\mathbf{x}\right| = \frac{1}{\|\mathbf{w}\|}\left|\mathbf{w}^\top\mathbf{x}_n + b - \mathbf{w}^\top\mathbf{x} - b\right| = \frac{1}{\|\mathbf{w}\|}$$

The SVM optimization problem

Maximize $\dfrac{1}{\|\mathbf{w}\|}$

subject to $\min\limits_{n=1,2,\ldots,N} |\mathbf{w}^\intercal \mathbf{x}_n + b| \;=\; 1$

Notice: $|\mathbf{w}^\intercal \mathbf{x}_n + b| \;= y_n \left(\mathbf{w}^\intercal \mathbf{x}_n + b\right)$

Minimize $\dfrac{1}{2}\, \mathbf{w}^\intercal \mathbf{w}$

subject to $y_n \left(\mathbf{w}^\intercal \mathbf{x}_n + b\right) \geq 1 \quad$ for $\quad n = 1, 2, \ldots, N$

Constrained optimization

Minimize $\frac{1}{2}\mathbf{w}^{\intercal}\mathbf{w}$

subject to $y_n\left(\mathbf{w}^{\intercal}\mathbf{x}_n + b\right) \geq 1$ for $n = 1, 2, \ldots, N$

$\mathbf{w} \in \mathbb{R}^d,\ b \in \mathbb{R}$

Lagrange? inequality constraints $\Longrightarrow$ KKT

Lagrange formulation

Minimize $\mathcal{L}(\mathbf{w}, b, \boldsymbol{\alpha}) = \frac{1}{2}\mathbf{w}^\intercal\mathbf{w} - \sum_{n=1}^{N} \alpha_n (y_n (\mathbf{w}^\intercal \mathbf{x}_n + b) - 1)$

w.r.t. $\mathbf{w}$ and b and maximize w.r.t. each $\alpha_n \geq 0$

$$\nabla_{\mathbf{w}}\mathcal{L} = \mathbf{w} - \sum_{n=1}^{N} \alpha_n y_n \mathbf{x}_n = \mathbf{0}$$

$$\frac{\partial \mathcal{L}}{\partial b} = -\sum_{n=1}^{N} \alpha_n y_n = 0$$

Dual problem formulation

$$\mathbf{w} = \sum_{n=1}^{N} \alpha_n y_n \mathbf{x}_n \qquad \text{and} \qquad \sum_{n=1}^{N} \alpha_n y_n = 0$$

in the Lagrangian

$$\mathcal{L}(\mathbf{w}, b, \boldsymbol{\alpha}) = \frac{1}{2}\mathbf{w}^{\intercal}\mathbf{w} - \sum_{n=1}^{N} \alpha_n \left(y_n \left(\mathbf{w}^{\intercal}\mathbf{x}_n + b\right) - 1\right)$$

we get

$$\mathcal{L}(\boldsymbol{\alpha}) = \sum_{n=1}^{N} \alpha_n - \frac{1}{2}\sum_{n=1}^{N}\sum_{m=1}^{N} y_n y_m \, \alpha_n \alpha_m \, \mathbf{x}_n^{\intercal}\mathbf{x}_m$$

Maximize w.r.t. to $\boldsymbol{\alpha}$ <u>subject to</u> $\alpha_n \geq 0$ for $n = 1, \cdots, N$ and $\sum_{n=1}^{N} \alpha_n y_n = 0$

It is a Quadratic Programming (QP) problem

$$\min_{\boldsymbol{\alpha}} \quad \frac{1}{2}\boldsymbol{\alpha}^{\mathsf{T}} \underbrace{\begin{bmatrix} y_1y_1\,\mathbf{x}_1^{\mathsf{T}}\mathbf{x}_1 & y_1y_2\,\mathbf{x}_1^{\mathsf{T}}\mathbf{x}_2 & \dots & y_1y_N\,\mathbf{x}_1^{\mathsf{T}}\mathbf{x}_N \\ y_2y_1\,\mathbf{x}_2^{\mathsf{T}}\mathbf{x}_1 & y_2y_2\,\mathbf{x}_2^{\mathsf{T}}\mathbf{x}_2 & \dots & y_2y_N\,\mathbf{x}_2^{\mathsf{T}}\mathbf{x}_N \\ \dots & \dots & \dots & \dots \\ y_Ny_1\,\mathbf{x}_N^{\mathsf{T}}\mathbf{x}_1 & y_Ny_2\,\mathbf{x}_N^{\mathsf{T}}\mathbf{x}_2 & \dots & y_Ny_N\,\mathbf{x}_N^{\mathsf{T}}\mathbf{x}_N \end{bmatrix}}_{\text{quadratic coefficients}} \boldsymbol{\alpha} + \underbrace{(-\mathbf{1}^{\mathsf{T}})}_{\text{linear}}\boldsymbol{\alpha}$$

subject to $\quad \underbrace{\mathbf{y}^{\mathsf{T}}\boldsymbol{\alpha} = 0}_{\text{linear constraint}}$

$$\underbrace{\mathbf{0}}_{\text{lower bounds}} \leq \boldsymbol{\alpha} \leq \underbrace{\infty}_{\text{upper bounds}}$$

LIBSVM: A **library** for **support vector machines**

CC Chang, CJ Lin - ACM transactions on intelligent systems and …, 2011 - dl.acm.org

… **LIBSVM** as a **library** for **support vector machines**.**LIBSVM** is currently one of the most widely used **SVM** … In this article,2 we present all implementation details of **LIBSVM**. However, this …

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Solution

Solution: $\boldsymbol{\alpha} = \alpha_1, \cdots, \alpha_N$

$$\Longrightarrow \quad \mathbf{w} = \sum_{n=1}^{N} \alpha_n y_n \mathbf{x}_n$$

KKT condition: For $n = 1, \cdots, N$

$$\alpha_n \left(y_n \left(\mathbf{w}^\intercal \mathbf{x}_n + b \right) - 1 \right) = 0$$

$\alpha_n > 0 \Longrightarrow \mathbf{x}_n$ is a $\boxed{\text{support vector}}$

Support vectors

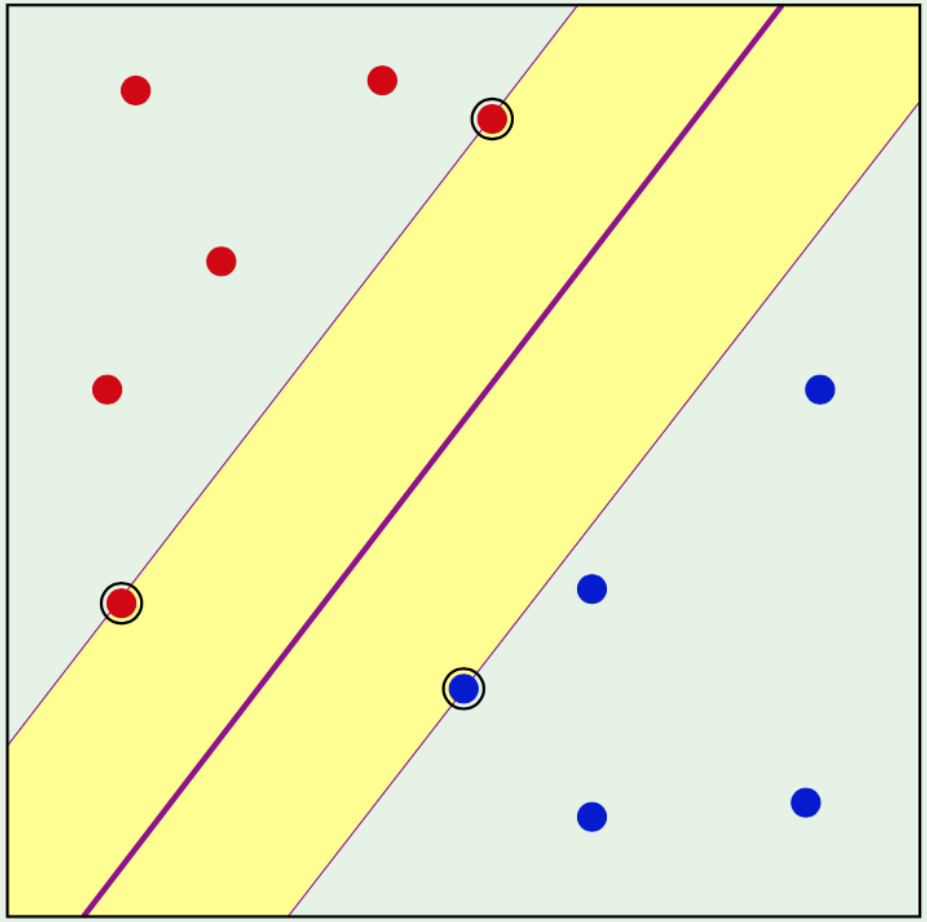
Closest $\mathbf{x}_n$'s to the plane: achieve the margin

$$\implies \quad y_n\left(\mathbf{w}^\intercal \mathbf{x}_n + b\right) = 1$$

$$\mathbf{w} \;=\; \sum_{\mathbf{x}_n \text{ is SV}} \alpha_n y_n \mathbf{x}_n$$

Solve for b using any SV:

$$y_n\left(\mathbf{w}^\intercal \mathbf{x}_n + b\right) = 1$$



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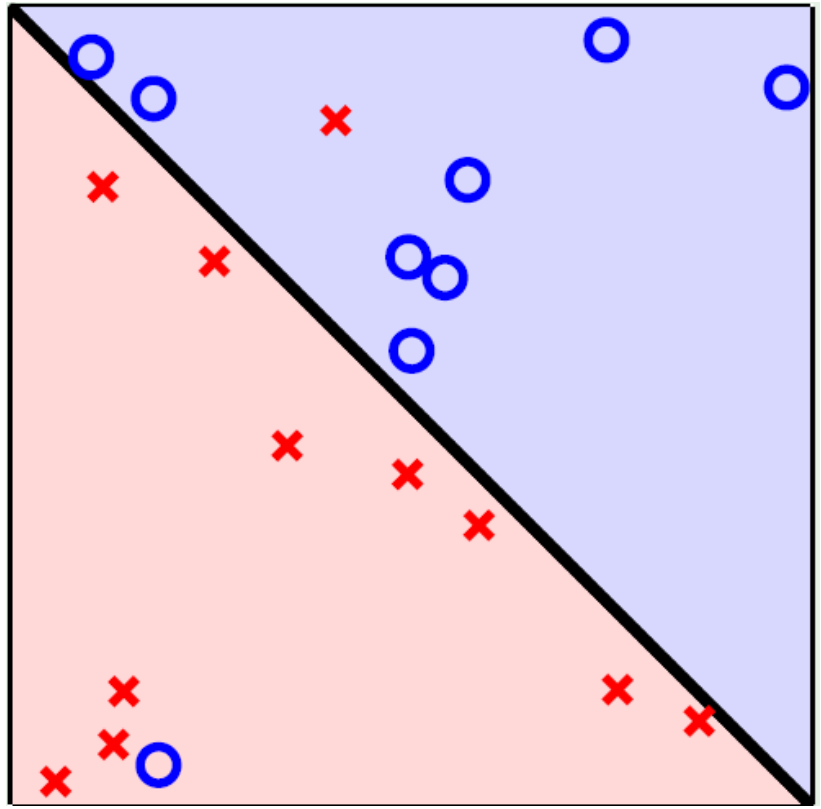
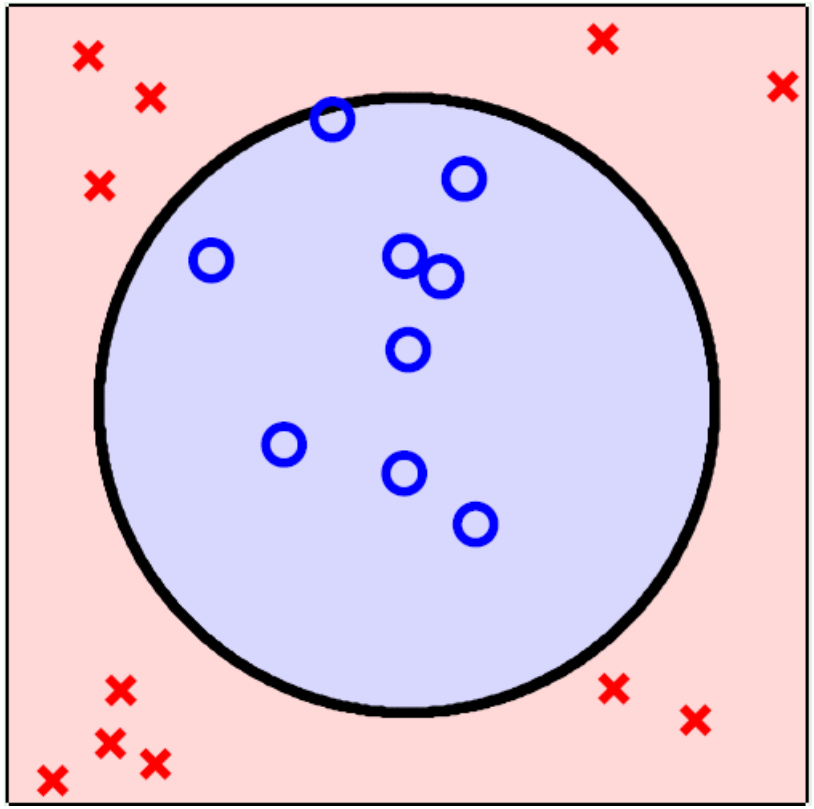
- Support Vector Machine (SVM)
  - history, main idea, mathematical derivation
- **Soft-margin SVM**
- Kernel methods

Non-separable Data



Slightly
non-separable



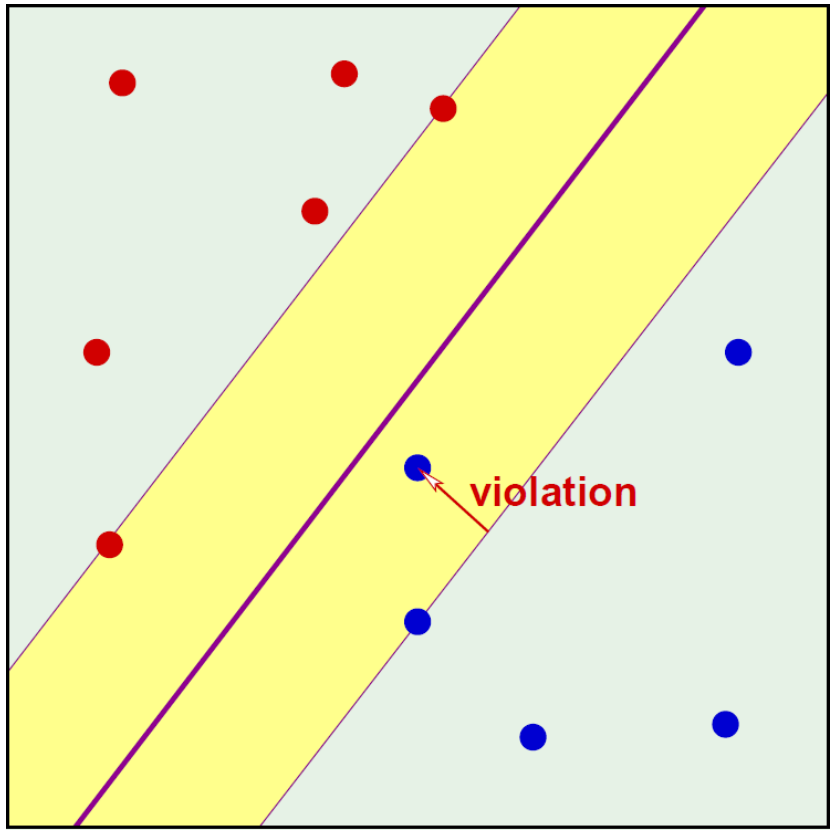
Seriously
non-separable

Soft-margin SVM

Margin violation: $y_n\left(\mathbf{w}^\intercal \mathbf{x}_n + b\right) \geq 1$ fails

Quantify: $y_n\left(\mathbf{w}^\intercal \mathbf{x}_n + b\right) \geq 1 - \xi_n$

$$\text{Total violation} = \sum_{n=1}^{N} \xi_n$$



ξ_n = ksi = “amount” of margin violation

$$\boxed{\xi_n \geq 0}$$

Soft-margin SVM optimization

- Similar to hard-margin optimization, but get compromise between maximizing the margin and allowing for violations

Still get large margin after term minimization

Allow for small violations by minimizing

$$\text{Minimize} \quad \frac{1}{2}\mathbf{w}^{\intercal}\mathbf{w} + C\sum_{n=1}^{N}\xi_n$$

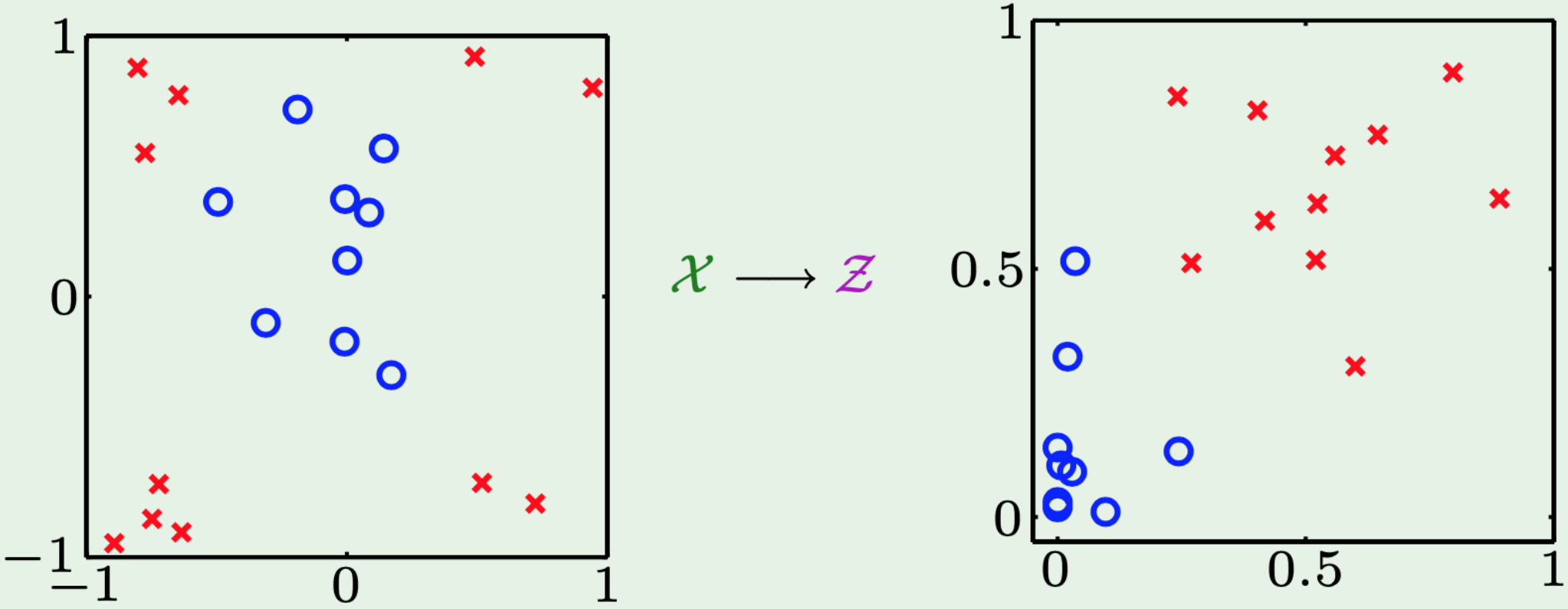
$$\text{subject to} \quad y_n\left(\mathbf{w}^{\intercal}\mathbf{x}_n + b\right) \geq 1 - \xi_n$$

$$\text{for} \quad n = 1, \ldots, N$$

$$\text{and} \quad \xi_n \geq 0 \quad \text{for} \quad n = 1, \ldots, N$$

Kernel transform

$$\mathcal{L}(\boldsymbol{\alpha}) = \sum_{n=1}^{N} \alpha_n - \frac{1}{2} \sum_{n=1}^{N} \sum_{m=1}^{N} y_n y_m \, \alpha_n \alpha_m \, \mathbf{z}_n^{\mathsf{T}} \mathbf{z}_m$$



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- **Kernel methods**

Computation from the Z space

$$\mathcal{L}(\boldsymbol{\alpha}) = \sum_{n=1}^{N} \alpha_n - \frac{1}{2} \sum_{n=1}^{N} \sum_{m=1}^{N} y_n y_m \, \alpha_n \alpha_m \, \mathbf{z}_n^\intercal \mathbf{z}_m$$

Constraints: $\alpha_n \geq 0$ for $n = 1, \cdots, N$ and $\sum_{n=1}^{N} \alpha_n y_n = 0$

$$\boxed{g(\mathbf{x}) = \text{sign}\left(\mathbf{w}^\intercal \mathbf{z} + b\right)}$$

need $\mathbf{z}_n^\intercal \mathbf{z}$

where $\mathbf{w} = \sum_{\mathbf{z}_n \text{ is SV}} \alpha_n y_n \mathbf{z}_n$

and b: $y_m \left(\mathbf{w}^\intercal \mathbf{z}_m + b\right) = 1$ need $\mathbf{z}_n^\intercal \mathbf{z}_m$

Generalized inner product

Given two points $\mathbf{x}$ and $\mathbf{x}' \in \mathcal{X}$, we need $\mathbf{z}^\intercal \mathbf{z}'$

Let $\mathbf{z}^\intercal \mathbf{z}' = K(\mathbf{x}, \mathbf{x}')$ (the kernel) "inner product" of $\mathbf{x}$ and $\mathbf{x}'$

Second order transform:

Example: $\mathbf{x} = (x_1, x_2) \longrightarrow$ 2nd-order Φ

$$\mathbf{z} = \Phi(\mathbf{x}) = (1, x_1, x_2, x_1^2, x_2^2, x_1 x_2)$$

$$K(\mathbf{x}, \mathbf{x}') = \mathbf{z}^\intercal \mathbf{z}' = 1 + x_1 x'_1 + x_2 x'_2 + x_1^2 x'^2_1 + x_2^2 x'^2_2 + x_1 x'_1 x_2 x'_2$$

Without transforming explicitly

- Can we compute $K(\boldsymbol{x}, \boldsymbol{x}')$ without transforming $\boldsymbol{x}$ and $\boldsymbol{x}'$?

Example:

$$K(\mathbf{x}, \mathbf{x}') = (1 + \mathbf{x}^\intercal \mathbf{x}')^2 = (1 + x_1 x'_1 + x_2 x'_2)^2$$

$$= 1 + x_1^2 x'^2_1 + x_2^2 x'^2_2 + 2x_1 x'_1 + 2x_2 x'_2 + 2x_1 x'_1 x_2 x'_2$$

Inner product

$$(1, x_1^2, x_2^2, \sqrt{2}x_1, \sqrt{2}x_2, \sqrt{2}x_1 x_2)$$

$$(1, x'^2_1, x'^2_2, \sqrt{2}x'_1, \sqrt{2}x'_2, \sqrt{2}x'_1 x'_2)$$

The polynomial kernel

$\mathcal{X} = \mathbb{R}^d$ and $\Phi : \mathcal{X} \to \mathcal{Z}$ is polynomial of order Q

The "equivalent" kernel $K(\mathbf{x}, \mathbf{x}') = (1 + \mathbf{x}^\intercal \mathbf{x}')^Q$

$$= (1 + x_1 x'_1 + x_2 x'_2 + \cdots + x_d x'_d)^Q$$

Adjust the scales: $K(\mathbf{x}, \mathbf{x}') = (a\mathbf{x}^\intercal \mathbf{x}' + b)^Q$

We only need Z to exist

- If $K(\boldsymbol{x}, \boldsymbol{x}')$ is an inner product in some space Z, that is ok.

- How do we know whether Z exists or not?
  1. By construction
  2. Math properties (Mercer's condition)
  3. Who cares?

For example,

$$K(\mathbf{x}, \mathbf{x}') = \exp\left(-\gamma \|\mathbf{x} - \mathbf{x}'\|^2\right)$$

Infinite-dimensional $\mathcal{Z}$: take simple case

$$K(x, x') = \exp\left(-(x - x')^2\right)$$

$$= \exp\left(-x^2\right) \exp\left(-x'^2\right) \underbrace{\sum_{k=0}^{\infty} \frac{2^k (x)^k (x')^k}{k!}}_{\exp(2xx')}$$

Kernel SVM

$$\min_{\boldsymbol{\alpha}} \quad \frac{1}{2}\boldsymbol{\alpha}^{\intercal}\underbrace{\begin{bmatrix} y_1y_1\,\mathbf{x}_1^{\intercal}\mathbf{x}_1 & y_1y_2\,\mathbf{x}_1^{\intercal}\mathbf{x}_2 & \dots & y_1y_N\,\mathbf{x}_1^{\intercal}\mathbf{x}_N \\ y_2y_1\,\mathbf{x}_2^{\intercal}\mathbf{x}_1 & y_2y_2\,\mathbf{x}_2^{\intercal}\mathbf{x}_2 & \dots & y_2y_N\,\mathbf{x}_2^{\intercal}\mathbf{x}_N \\ \dots & \dots & \dots & \dots \\ y_Ny_1\,\mathbf{x}_N^{\intercal}\mathbf{x}_1 & y_Ny_2\,\mathbf{x}_N^{\intercal}\mathbf{x}_2 & \dots & y_Ny_N\,\mathbf{x}_N^{\intercal}\mathbf{x}_N \end{bmatrix}}_{\text{quadratic coefficients}}\boldsymbol{\alpha} + \underbrace{(-\mathbf{1}^{\intercal})}_{\text{linear}}\boldsymbol{\alpha}$$

$$\text{subject to} \quad \underbrace{\mathbf{y}^{\intercal}\boldsymbol{\alpha} = 0}_{\text{linear constraint}}$$

$$\underbrace{\mathbf{0}}_{\text{lower bounds}} \leq \boldsymbol{\alpha} \leq \underbrace{\infty}_{\text{upper bounds}}$$

Kernel formulation

$$\underbrace{\begin{bmatrix} y_1y_1K(x_1,x_1) & y_1y_2K(x_1,x_2) & \dots & y_1y_NK(x_1,x_N) \\ y_2y_1K(x_2,x_1) & y_2y_2K(x_2,x_2) & \dots & y_2y_NK(x_2,x_N) \\ \dots & \dots & \dots & \dots \\ y_Ny_1K(x_N,x_1) & y_Ny_2K(x_N,x_2) & \dots & y_Ny_NK(x_N,x_N) \end{bmatrix}}_{\text{quadratic coefficients}}$$

Solution to kernel SVM

Express $g(\mathbf{x}) = \text{sign}\left(\mathbf{w}^\intercal \mathbf{z} + b\right)$ in terms of $K(-,-)$

$$\mathbf{w} = \sum_{\mathbf{z}_n \text{ is SV}} \alpha_n y_n \mathbf{z}_n \implies g(\mathbf{x}) = \text{sign}\left(\sum_{\alpha_n > 0} \alpha_n y_n K(\mathbf{x}_n, \mathbf{x}) + b\right)$$

where $b = y_m - \sum_{\alpha_n > 0} \alpha_n y_n K(\mathbf{x}_n, \mathbf{x}_m)$

for any support vector $(\alpha_m > 0)$

Mercer's condition

$K(\mathbf{x}, \mathbf{x}')$ is a valid kernel iff

1. It is symmetric and 2. The matrix:

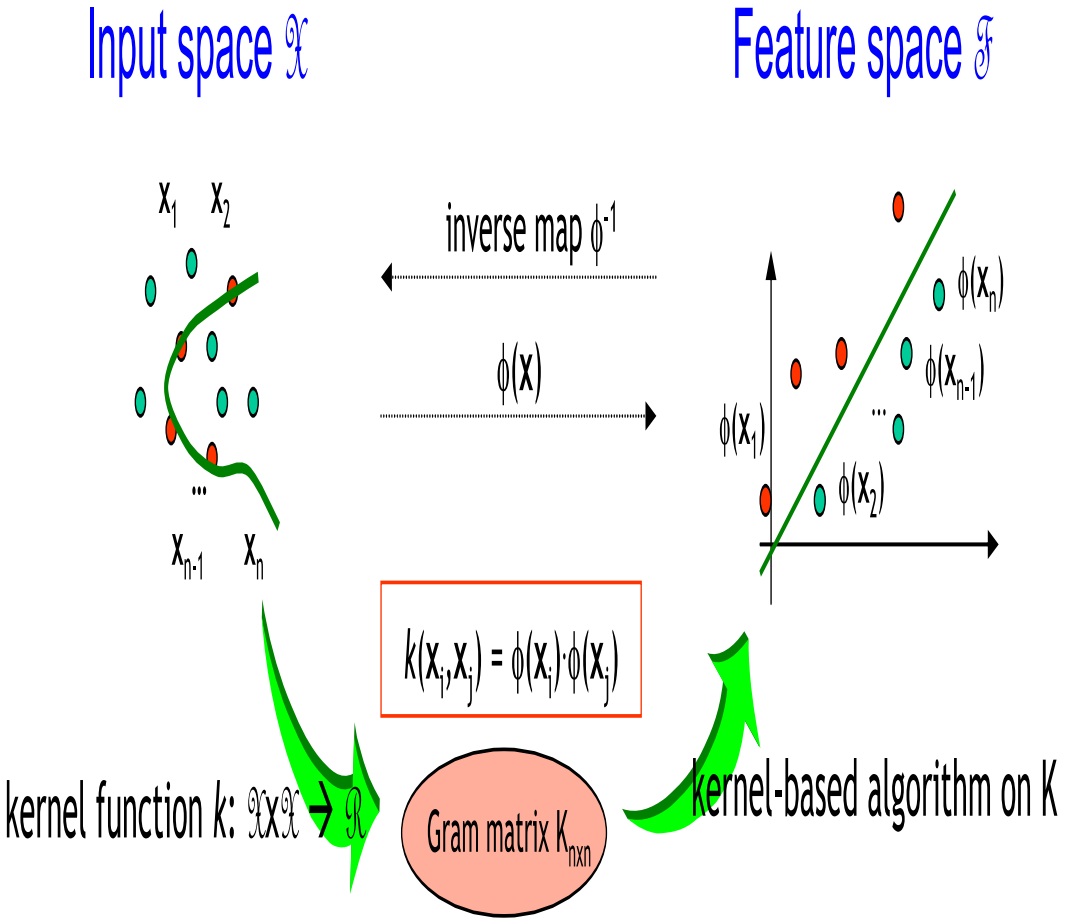
$$\begin{bmatrix} K(\mathbf{x}_1, \mathbf{x}_1) & K(\mathbf{x}_1, \mathbf{x}_2) & \dots & K(\mathbf{x}_1, \mathbf{x}_N) \\ K(\mathbf{x}_2, \mathbf{x}_1) & K(\mathbf{x}_2, \mathbf{x}_2) & \dots & K(\mathbf{x}_2, \mathbf{x}_N) \\ \dots & \dots & \dots & \dots \\ K(\mathbf{x}_N, \mathbf{x}_1) & K(\mathbf{x}_N, \mathbf{x}_2) & \dots & K(\mathbf{x}_N, \mathbf{x}_N) \end{bmatrix}$$

is **positive semi-definite**

for any $\mathbf{x}_1, \cdots, \mathbf{x}_N$ (Mercer's condition)

Then, you can design your own kernel as above.

Summary



- Map the data from $\mathcal{X}$ into a (high-dimensional) vector space, the feature space $\mathcal{F}$, by applying the feature map ϕ on the data points **x**.
- Find a linear (or other easy) pattern in $\mathcal{F}$ using a well-known algorithm (that works on the Gram matrix).
- By applying the inverse map, the linear pattern in $\mathcal{F}$ can be found to correspond to a complex pattern in $\mathcal{X}$.
- This implicitly by only making use of inner products in $\mathcal{F}$ (kernel trick)

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